NARUTATSU (EDWARD) RI

605-615 W 115th St, New York, NY 10027 | 917-244-2814 | wl2787@columbia.edu

EDUCATION

Columbia University, The Fu Foundation School of Engineering and Applied Science **Sep. 2020 – May 2024**
Egleston Scholar, Top 1% of Columbia Engineering Applicants
Bachelor of Science – Applied Mathematics, Computer Science Minor, GPA: 4.06/4
Relevant Coursework: Fundamentals of Computer Systems, Multivariable Calculus, Data Structures, Unsupervised Learning

Tsukuba University **Apr. 2020 – Sep. 2020**
Bachelor of Science – Information Science
Obtained full marks on mathematics section entrance exam, attended for six months before matriculating to Columbia Engineering

Seiko Gakuin High School **Apr. 2017 – Mar. 2020**
Valedictorian – Science Division, GPA: 5.0/5.0
Seiko Gakuin (聖光学院中学校高等学校), known as one of the most prestigious educational institutions in Japan, requires students to pass an entrance exam in the sixth grade to gain admission

SKILLS

Computer: Python, C/C++, Java, MATLAB, Linux/Unix Systems, HTML/CSS
Language: Japanese, English, Chinese, Russian (Elementary proficiency)
Interests: Machine Learning, Mathematics

RESEARCH/TECHNICAL EXPERIENCE

Department of Computer Science, Columbia University **Jan. 2021 – Present**
Independent Researcher with Professor Nakul Verma

- Currently working on developing a new method for word embeddings
- Conducted mathematical analysis on skip-gram model
- Participating in weekly one-on-one meetings with professor, reading multiple mathematically rich papers on a meeting basis, evaluating potential new ideas through writing custom optimizers and cost functions in Python and validating them
- Previously worked on unsupervised sentiment analysis on stock articles and devised new bootstrapping method for accuracy improvement

Thryving, DATAFULCT, INC. **Jul. 2021 – Present**
Software Engineer (Subcontractor)

- Developing an interactive business intelligence provider responsible for the backend processing and visualization generations
- Working on releasing an API of the business intelligence provider

iSYSLab, University of Electro-Communications **Oct. 2017 – Jun. 2019**
Special Researcher with Professor Takayuki Nagai

- Approached professor through email with research idea regarding dialogue systems in 10th grade and passed special entrance laboratory examination
- Conducted research for 2.5 years under Dr. Takayuki Nagai (currently Professor at University of Osaka)
- Presented research at The 33rd Annual Conference of the Japanese Society for Artificial Intelligence, 2019 (JSAI 2019): "A Dialogue System Implemented with Latent Parameters" as first high school participant accepted for publication

LEADERSHIP ACTIVITIES

Columbia University Formula Racing **Sep. 2020 – Jan. 2021**
Software Team, Tire Simulation Lead

- Led optimization group
- Conducted tire optimization and analysis with MATLAB utilizing Pacejka's Magic Formula

Columbia Space Initiative **Sep. 2020 – Present**
RASC-AL Team

- Developed plan for Venus flyby mission
- Designed propulsion systems

Coronavirus Visualization Team **Apr. 2020 – Sep. 2020**
Mental Health Team, Data Analyst

- Scraped and generated visualizations of relationships between quarantine and depression/anxiety cases
- Generated map for available healthcare facilities around the United States

AWARDS

Egleston Scholar **Apr. 2020**

- Selected as one of nine students from Columbia Engineering's class of 2024

Full Ride Scholarship, Ezoe Memorial Foundation **Dec. 2019**

- 49th year Academic Scholar
- Selected as one of few undergraduate students in Japan awarded a full-ride scholarship (annual, $92880)

Winner, RoboCup Japan Open **May 2018**

- Built and trained deep learning model for translating natural language to robot commands alone

Second Place, PC Koshien (National Programming Contest) **Oct. 2017**

Finalist, S-ISEF (Research on Space Exploration) **Nov. 2018**

- Presented swarm technology-based idea to officials from RIKEN and Astronauts